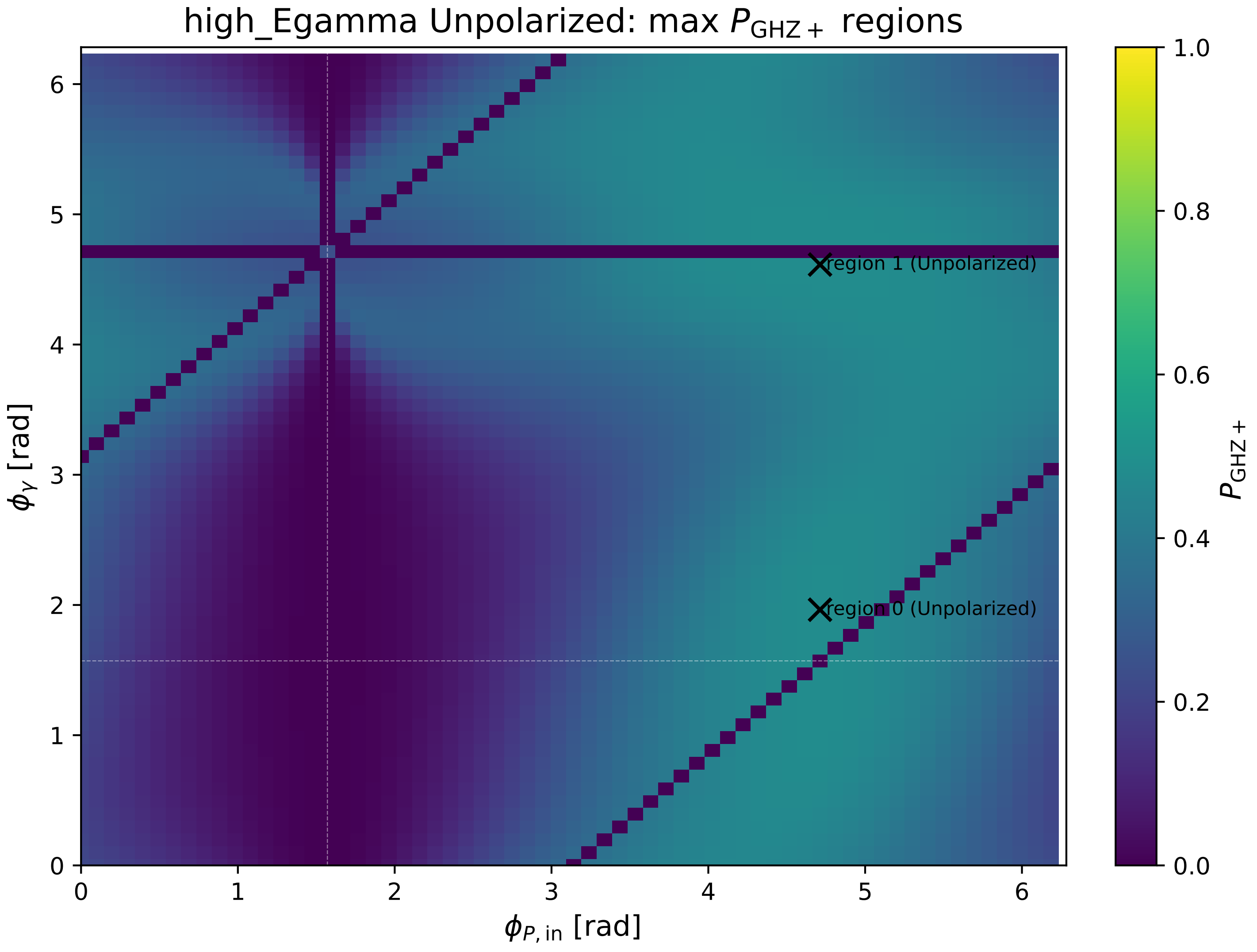
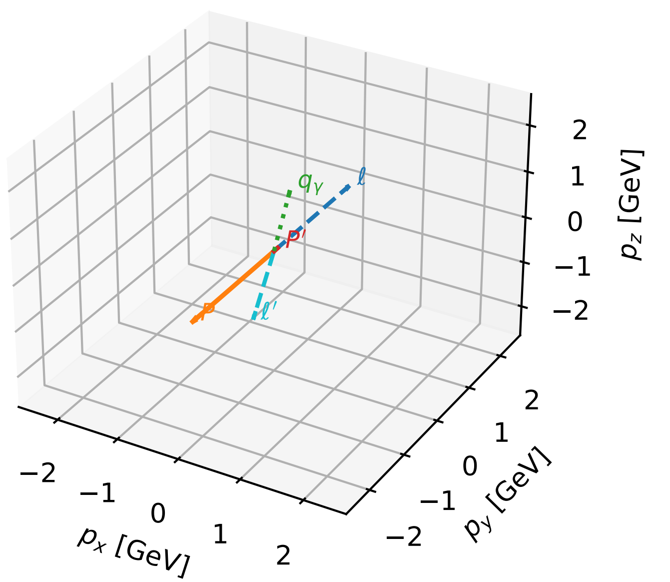


high_Egamma Unpolarized: max $P_{\mathrm{GHZ}+}$ regions
region 1 (Unpolarized)
region 0 (Unpolarized)
ϕ_γ [rad]
$\phi_{P,\mathrm{in}}$ [rad]
$P_{\mathrm{GHZ}+}$
0
1
2
3
4
5
6
0.0
0.2
0.4
0.6
0.8
1.0

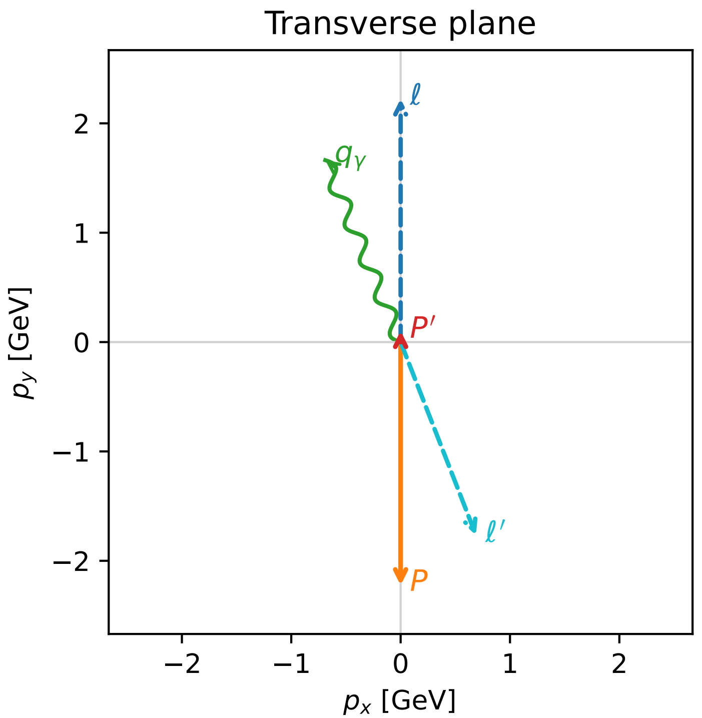
Unpolarized characteristic max $P_{\mathrm{GHZ}+}$ configuration

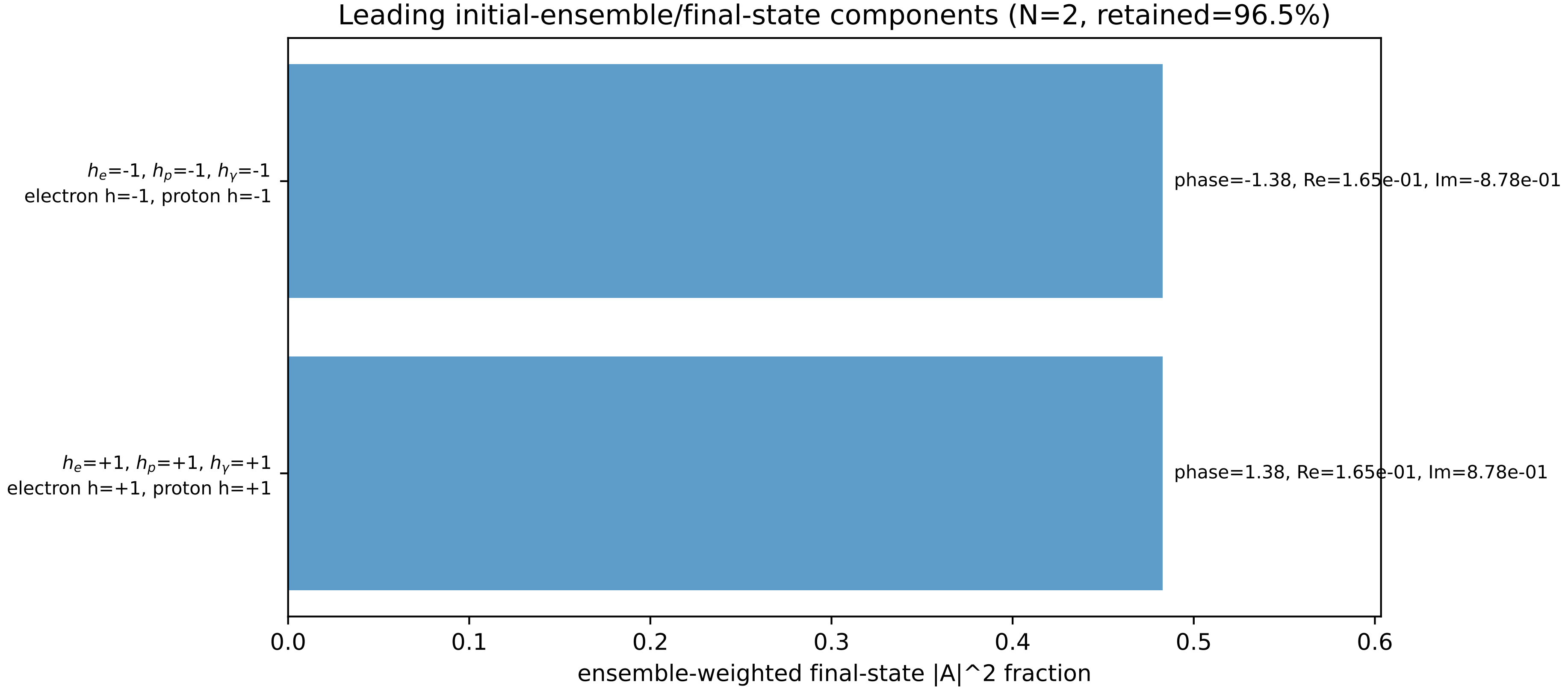



```
ghz_purity_high_egamma_unpolarized_region_0  P_{\rm GHZ+}=0.482724
region: high_Egamma_unpolarized_region_0
outgoing-state purity: 0.498924
kinematic point: high_s_high_theta_in_high_Egamma
```
incoming state: $\frac{1}{4}\sum_{h_e,h_p}\rho(h_e,h_p)$

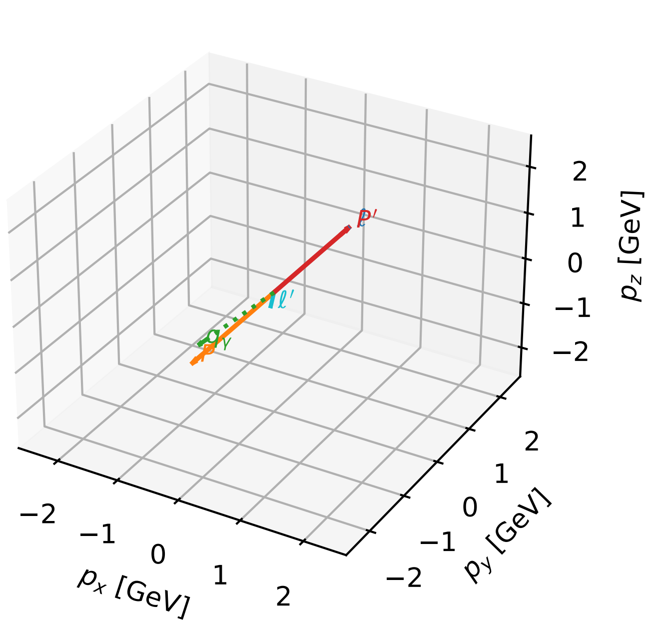
s=21.5158, $\sqrt{s}$=4.63852, $|\vec{P}|$=2.22442, $|\vec{P}'|$=0.102337
θ_{in}=1.57079, ϕ_ℓ=1.5708, ϕ_P=4.71239
E_γ=1.8, ϕ_γ=1.9635
Q^2=16.2839, x_B=0.841965, t=-3.25132, W^2=3.93631, y=0.937217
$\theta(\ell',\gamma)$=178.816 deg

four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.2244 p_x= 0.0000 p_y= 2.2244 p_z= -0.0000
P E= 2.4141 p_x= -0.0000 p_y= -2.2244 p_z= 0.0000
ℓ' E= 1.8950 p_x= 0.6888 p_y= -1.7653 p_z= 0.0000
P' E= 0.9436 p_x= 0.0000 p_y= 0.1023 p_z= 0.0000
q_γ E= 1.8000 p_x= -0.6888 p_y= 1.6630 p_z= 0.0000





Unpolarized characteristic max $P_{\mathrm{GHZ}+}$ configuration




```
ghz_purity_high_egamma_unpolarized_region_1  P_{\rm GHZ+}=0.482457
region: high_Egamma_unpolarized_region_1
outgoing-state purity: 0.499933
kinematic point: high_s_high_theta_in_high_Egamma
```

incoming state: $\frac{1}{4}\sum_{h_e,h_p}\rho(h_e,h_p)$

s=21.5158, $\sqrt{s}$=4.63852, $|\vec{P}|$=2.22442, $|\vec{P}'|$=2.20096
$\theta_{\rm in}$=1.57079, ϕ_ℓ=1.5708, ϕ_P=4.71239
E_γ=1.8, ϕ_γ=4.61421
Q^2=3.80661, x_B=0.187472, t=-19.5835, W^2=17.3782, y=0.983958
$\theta(\ell',\gamma)$=28.9269 deg

four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.2244 p_x= 0.0000 p_y= 2.2244 p_z= -0.0000
P E= 2.4141 p_x= -0.0000 p_y= -2.2244 p_z= 0.0000
ℓ' E= 0.4460 p_x= 0.1764 p_y= -0.4096 p_z= 0.0000
P' E= 2.3925 p_x= 0.0000 p_y= 2.2010 p_z= 0.0000
q_γ E= 1.8000 p_x= -0.1764 p_y= -1.7913 p_z= 0.0000